

Last time info. updated: 03-08-2026, where p_beat_e190: 0.0519, p_beat737: 0.1634 and p_non_arrival= 0.7847

Number of possible states: 56

Possible states generated OK.-

Number of reachable states: 39. Hence, 17 states were discarded.

Reachable states are:

0	0	0	0	0
0	0	1	0	0
0	0	0	0	1
0	1	0	0	0
0	1	1	0	0
0	1	0	0	1
0	0	0	1	0
0	0	1	1	0
0	0	0	1	1
1	0	0	0	0
1	0	1	0	0
1	0	0	0	1
1	1	0	0	0
1	1	1	0	0
1	1	0	0	1
1	0	0	1	0
1	0	1	1	0
1	0	0	1	1
0	0	2	0	0
0	0	1	0	1
0	1	2	0	0
0	1	1	0	1
0	0	2	1	0
0	0	1	1	1
0	2	0	0	0
0	2	1	0	0
0	2	0	0	1
0	1	0	1	0
0	1	1	1	0
0	1	0	1	1
1	2	0	0	0
1	1	0	1	0
2	0	0	0	0
2	0	1	0	0
2	0	0	0	1
2	1	0	0	0
2	0	0	1	0
1	0	2	0	0
1	0	1	0	1

-----P1 RUN OK-----

Transition matrix (P), with arrival probs. = Markov matrix generated OK.-

Row sums (should all be 1, otherwise check that row's probs):

Columns 1 through 20

	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1																

Columns 21 through 39

```
      1      1      1      1      1      1      1      1      1      1      1      1      1      1      1      1
1      1      1
```

-----P2 RUN OK-----

Eigenvalues:

```
1.0000 + 0.0000i
-0.2179 - 0.2446i
-0.2179 + 0.2446i
-0.2816 - 0.1141i
-0.2816 + 0.1141i
0.1734 - 0.3118i
0.1734 + 0.3118i
0.2844 + 0.0000i
0.1022 - 0.2439i
0.1022 + 0.2439i
0.0002 + 0.0000i
0.0001 - 0.0002i
0.0001 + 0.0002i
-0.0002 - 0.0001i
-0.0002 + 0.0001i
0.0000 - 0.0000i
0.0000 + 0.0000i
-0.0000 - 0.0000i
-0.0000 + 0.0000i
0.0000 - 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
-0.0000 + 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
-0.0000 - 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i
-0.0000 - 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
0.0000 - 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i
```

Number of eigenvalues equal to 1: 1

Selected eigenvector column: 1

Double checking ($v * P = v$) Difference (should be close to 0): 0.0000000000

Used tolerance: 1.0000e-05

-> v IS the stationary distribution

Theoretical stationary distribution:

Columns 1 through 12

0.3403	0.0932	0.0709	0.0888	0.0242	0.0185	0.0709	0.0193	0.0148
0.0890	0.0197	0.0185						

Columns 13 through 24

0.0233	0.0055	0.0049	0.0183	0.0041	0.0038	0.0047	0.0147	0.0012
0.0038	0.0010	0.0030						

Columns 25 through 36

0.0044	0.0013	0.0009	0.0138	0.0039	0.0029	0.0012	0.0038	0.0044
0.0003	0.0009	0.0013						

Columns 37 through 39

0.0009	0.0009	0.0029
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-----P3 RUN OK-----

For h=5; capacity=3:

_____SIMULATION 1_____

Starting from State (p0(1)):

Set starting state: 1

Visit counts set to 0.-

RESULTS - From State 1 [0 0 0 0] (empty):

After 1000000 beats:

-e190:

Arrivals: 51525

Effectively seated: 50615

Rejected: 910

-737:

Arrivals: 162859

Effectively seated: 160043

Rejected: 2816

-Total:

Arrivals: 214384

Serviced: 210658

Rejected: 3726

Per beat:

-e190:

Arrivals 0.0515

Effectively seated 0.0506

Rejected: 0.0009

-737:

Arrivals 0.1629

Effectively seated 0.1600

Rejected: 0.0028

-Total:
 Arrivals 0.2144
 Effectively seated 0.2107
 Rejected: 0.0037

Rejection rate total: 1.7380%
 Rejection rate 737: 1.7291%
 Rejection rate e190: 1.7661%

SIMULATION 2

Starting from State (p0(37)):
 Set starting state: 37

Visit counts set to 0.-

RESULTS - From State 39 [2 0 0 1 0] (full):
 After 1000000 beats:

-e190:
 Arrivals: 52236
 Effectively seated: 51275
 Rejected: 961

-737:
 Arrivals: 163615
 Effectively seated: 160644
 Rejected: 2971

-Total:
 Arrivals: 215851
 Serviced: 211919
 Rejected: 3932

Per beat:

-e190:
 Arrivals 0.0522
 Effectively seated 0.0513
 Rejected: 0.0010

-737:
 Arrivals 0.1636
 Effectively seated 0.1606
 Rejected: 0.0030

-Total:
 Arrivals 0.2159
 Effectively seated 0.2119
 Rejected: 0.0039

Rejection rate total: 1.8216%
 Rejection rate 737: 1.8158%
 Rejection rate e190: 1.8397%

RSS DIFF. VALUES FOR MY REPORT		
Step	Empty start	Full start
10	3.903078e-02	1.995955e-01
1000	6.443794e-04	3.526802e-04

100000	3.572274e-06	1.543573e-05
1000000	1.310726e-06	1.098112e-06

RANKING OF STATES

Rank	State	Theory	Empty start	Full start	Cum. theory	Cum. empty	Cum. full
1	00000	34.0278	34.1126	33.9376	34.0278	34.1126	33.9376
2	00100	9.3201	9.3394	9.3075	43.3479	43.4520	43.2451
3	10000	8.8970	8.9274	8.8907	52.2449	52.3794	52.1358
4	01000	8.8834	8.8923	8.8812	61.1283	61.2717	61.0170
5	00010	7.0857	7.0808	7.0860	68.2140	68.3525	68.1030
6	00001	7.0857	7.1140	7.0950	75.2997	75.4665	75.1980
7	01100	2.4195	2.4278	2.4457	77.7192	77.8943	77.6437
8	11000	2.3293	2.3268	2.3522	80.0485	80.2211	79.9959
9	10100	1.9716	1.9611	1.9788	82.0201	82.1822	81.9747
10	00110	1.9258	1.9394	1.9484	83.9459	84.1216	83.9231

-----P4 RUN OK-----

Theoretical long-run occupancy shares (from v):

P(all bays empty) = 0.3403
 P(1 bay busy) = 0.4127
 P(2 bays busy) = 0.1986
 P(full) = 0.0485
 Check (should sum to 1): 1.0000
 Mean number of busy bays = 0.9552 of 3
 Long-run utilisation = 31.84%

The expected parameters in the theoretical long run p/plane-p/beat are as follows:

Note: arrival rates were estimated from ADL Airport data using Python.

Aircraft	Arrivals	Rejected	Serviced
E190	0.0519	0.0009	0.0510
737	0.1634	0.0029	0.1605
Total	0.2153	0.0039	0.2114

The expected parameters in the theoretical long run p/plane-p/day are as follows:

Aircraft	Arrivals	Rejected	Serviced
E190	12.4560	0.2244	12.2316
737	39.2160	0.7065	38.5095
Total	51.6720	0.9309	50.7411

-----P5 RUN OK-----

Plot generated OK.-

-----P6 RUN OK-----

_____Economic Layer_____

Service team cost p/hour: 398.7100 AUD

Complete team cost p/beat: 39.8710 AUD

e190:

Service time: 3 beats
 Service time: 18 minutes
 Estimated staff cost: 119.6130 AUD

737:

Service time: 5 beats
 Service time: 30 minutes

Estimated staff cost: 199.3550 AUD

Net profit per aircraft:

e190: 1323.9000 - 119.6130 = 1204.2870 AUD, Contribution p/beat 401.4290 AUD

737: 2471.2800 - 199.3550 = 2271.9250 AUD, Contribution p/beat 454.3850 AUD

In terms of net contribution: The 737 is worth 13.19% p/beat p/bay than the e190.

-----P7 RUN OK-----						
Multiplier	Reject %	e190 arrivals	e190 accepted	e190 rejected	737 arrivals	737 accepted
737 rejected						
0.00	0.00	0.000	0.000	0.000	0.000	
0.000	0.000					
0.10	0.00	1.246	1.246	0.000	3.922	
3.921	0.000					
0.20	0.02	2.491	2.491	0.000	7.843	
7.842	0.002					
0.30	0.06	3.737	3.734	0.002	11.765	
11.757	0.008					
0.40	0.15	4.982	4.975	0.007	15.686	
15.663	0.023					
0.50	0.28	6.228	6.211	0.017	19.608	
19.554	0.054					
0.60	0.46	7.474	7.439	0.034	23.530	
23.422	0.108					
0.70	0.70	8.719	8.658	0.061	27.451	
27.259	0.192					
0.80	1.00	9.965	9.865	0.100	31.373	
31.058	0.315					
0.90	1.37	11.210	11.057	0.154	35.294	
34.811	0.484					
1.00	1.80	12.456	12.232	0.224	39.216	
38.509	0.707					
1.10	2.30	13.702	13.387	0.315	43.138	
42.147	0.991					
1.20	2.85	14.947	14.520	0.427	47.059	
45.716	1.343					
1.30	3.47	16.193	15.630	0.562	50.981	
49.210	1.771					
1.40	4.15	17.438	16.715	0.724	54.902	
52.625	2.278					
1.50	4.88	18.684	17.772	0.912	58.824	
55.954	2.870					
1.60	5.66	19.930	18.802	1.128	62.746	
59.195	3.551					
1.70	6.48	21.175	19.802	1.373	66.667	
62.344	4.323					
1.80	7.35	22.421	20.772	1.649	70.589	
65.398	5.190					
1.90	8.26	23.666	21.712	1.955	74.510	
68.357	6.154					
2.00	9.20	24.912	22.621	2.291	78.432	
71.218	7.214					
2.10	10.17	26.158	23.499	2.659	82.354	
73.982	8.372					

2.20	11.16	27.403	24.345	3.058	86.275	
76.648	9.627					
2.30	12.17	28.649	25.162	3.487	90.197	
79.218	10.979					
2.40	13.20	29.894	25.947	3.947	94.118	
81.692	12.426					
2.50	14.25	31.140	26.703	4.437	98.040	
84.072	13.968					
Multiplier	Reject %	Captured AUD	Missed e190	Missed 737	Missed AUD	Potential AUD
0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.10	0.00	10409.37	0.04	0.23	0.27	10409.64
0.20	0.02	20815.15	0.60	3.53	4.13	20819.28
0.30	0.06	31208.78	2.90	17.24	20.14	31228.92
0.40	0.15	41577.28	8.83	52.45	61.28	41638.56
0.50	0.28	51904.29	20.74	123.17	143.91	52048.20
0.60	0.46	62171.02	41.33	245.50	286.83	62457.85
0.70	0.70	72357.12	73.55	436.82	510.37	72867.49
0.80	1.00	82441.50	120.42	715.21	835.63	83277.13
0.90	1.37	92403.02	184.99	1098.76	1283.75	93686.77
1.00	1.80	102221.03	270.25	1605.13	1875.38	104096.41
1.10	2.30	111875.94	379.01	2251.11	2630.11	114506.05
1.20	2.85	121349.52	513.90	3052.27	3566.17	124915.69
1.30	3.47	130625.27	677.29	4022.77	4700.06	135325.33
1.40	4.15	139688.55	871.31	5175.11	6046.42	145734.97
1.50	4.88	148526.77	1097.75	6520.09	7617.85	156144.61
1.60	5.66	157129.37	1358.15	8066.73	9424.88	166554.26
1.70	6.48	165487.90	1653.72	9822.27	11475.99	176963.90
1.80	7.35	173595.90	1985.40	11792.24	13777.64	187373.54
1.90	8.26	181448.81	2353.83	13980.54	16334.37	197783.18
2.00	9.20	189043.91	2759.41	16389.50	19148.91	208192.82
2.10	10.17	196380.09	3202.31	19020.06	22222.37	218602.46
2.20	11.16	203457.80	3682.45	21871.85	25554.30	229012.10
2.30	12.17	210278.80	4199.58	24943.36	29142.94	239421.74
2.40	13.20	216846.06	4753.28	28232.04	32985.33	249831.38
2.50	14.25	223163.55	5342.97	31734.50	37077.47	260241.02

SUMMARY - VALUES FOR MY REPORT

	Theory	Simulation 1 (empty)	Simulation 2(full)
e190 arrivals	0.05190	0.05153	0.05224
e190 serviced	0.05096	0.05062	0.05128
e190 rejected	0.00094	0.00091	0.00096
737 arrivals	0.16340	0.16286	0.16362
737 serviced	0.16046	0.16004	0.16064
737 rejected	0.00294	0.00282	0.00297
TOTAL arrivals	0.21530	0.21438	0.21585
TOTAL serviced	0.21142	0.21066	0.21192
TOTAL rejected	0.00388	0.00373	0.00393

-----P9 RUN OK-----

Correctly saved

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